

**PEAK ACHIEVERS
ONLINE ACADEMY**

INTEGRATED SCIENCE

GRADE 8

**BOOK 1
CBE LEARNER & REVISION EDITION**

CONCISE NOTES • PRACTICALS • CBE ACTIVITIES
100 MIXED CONCEPTS • 5 MODEL TESTS

LEARN • APPLY • THINK BIG • ACHIEVE

HEZEKIAH OMWERI NYATUKA

B.Ed. (Science) - Chemistry & Physics
Kenyatta University

Copyright

Peak Achievers Integrated Science - Grade 8, Book 1

CBE Learner & Revision Edition

Copyright © 2026 Hezekiah Omweri Nyatuka. All rights reserved.

No part of this publication may be reproduced or distributed without permission except for reasonable educational use permitted by applicable law.

This book has been developed as an independent educational and revision resource. It is not an official examination paper or publication of KNEC, KICD or the Ministry of Education.

Peak Achievers Online Academy

Email: hezekiahomweri@gmail.com

Phone: 0740166105 / 0740355410

Website: www.peakachievers.co.ke

Dedication

In loving memory of my late father, Simeon Nyatuka, whose guidance, sacrifice and inspiration remain part of my educational journey.

With heartfelt appreciation to Hyrine Kerubo, Peter Munde Mong'are, Teresa Kwamboka Sawe, Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob, Tim, Mr Peter Kimani, and all those whose support and encouragement have strengthened this journey.

To Her, my special one, for being a special part of my journey and for the place you hold in my heart.



To my fellow teachers, guardians, family and every learner who continues to inspire the work behind Peak Achievers.

May every learner using this book remain curious, work with determination, and discover the confidence to learn, apply, think big and achieve.

LEARN • APPLY • THINK BIG • ACHIEVE

About the Author

Hezekiah Omweri Nyatuka is an educator, online tutor and educational content developer with a Bachelor of Education (Science) degree, specialising in Chemistry and Physics, from Kenyatta University.

His educational work focuses on helping learners understand scientific ideas through clear explanations, practical activities, real-life applications, problem-solving and competency-based learning.

Through Peak Achievers Online Academy, he supports learners through educational resources, revision materials and tutoring in Mathematics and the sciences.

AUTHOR MESSAGE Science becomes meaningful when we connect what we learn in class with what happens around us every day. Keep asking questions. Keep investigating. Keep practising. Keep thinking big.

How to Use This Book

Each sub-strand combines concise notes with examples, diagrams, practical activities, real-life applications and CBE-style assessment.

Feature	Purpose
REMEMBER	Essential concept to retain.
HINT	Guidance for solving or understanding.
SPOTLIGHT	Important extension or connection.
CAUTION / ERROR DETECTIVE	Common misconception or safety point.
THINK BIG	Higher-order reasoning and application.
EXAM SMART	Assessment strategy and precision.

Recommended learning sequence:

REVISE → PRACTISE → APPLY → THINK BIG → ASSESS → CHECK → ACHIEVE

Table of Contents

- STRAND 1: Mixtures, Elements and Compounds
 - 1.1 Elements and Compounds
 - 1.2 Physical and Chemical Changes
 - 1.3 Classes of Fire
- Strand 1 Revision
- STRAND 2: Living Things and their Environment
 - 2.1 The Cell
 - 2.2 Movement of Materials In and Out of the Cell
 - 2.3 Reproduction in Human Beings
- Strand 2 Revision
- STRAND 3: Force and Energy
 - 3.1 Transformation of Energy
 - 3.2 Pressure
- Strand 3 Revision
- Comprehensive Revision - 100 Mixed Concepts
- Learner Progress Checklist
- Final CBE National-Exam-Style Model Tests 1-5
- Answers and Marking Guides

STRAND 1: MIXTURES, ELEMENTS AND COMPOUNDS

1.1 Elements and Compounds

Matter is anything that has mass and occupies space. At Grade 8, we use particle ideas to distinguish elements, compounds and mixtures.

Atoms, Elements and Chemical Symbols

An atom is a basic particle of an element. An element is a pure substance made of one type of atom and cannot be broken down into simpler substances by ordinary chemical means.

Element	Symbol	Element	Symbol
Hydrogen	H	Oxygen	O
Carbon	C	Nitrogen	N
Sodium	Na	Chlorine	Cl
Iron	Fe	Copper	Cu
Aluminium	Al	Silver	Ag
Gold	Au	Mercury	Hg
Lead	Pb		

CAUTION Chemical symbols are case-sensitive: Cl is chlorine, not CL; Na is sodium, not NA.

Molecules and Compounds

A molecule consists of two or more atoms chemically bonded. A molecule may contain atoms of the same element, for example O₂, or different elements, for example H₂O.

A compound is a pure substance formed when atoms of different elements combine chemically in fixed proportions.

REMEMBER Not every molecule is a compound. O₂ is a molecule but not a compound because it contains only oxygen atoms.

Word Equations

Word equations show reactants on the left and products on the right.

Carbon + Oxygen → Carbon dioxide

The starting substances are reactants; the substances formed are products.

Elements, Compounds and Mixtures

Feature	Element	Compound	Mixture
Composition	One type of atom	Different elements chemically combined	Substances physically combined
Proportion	Fixed	Fixed	Variable

Separation	Not by ordinary chemical breakdown into simpler substances	Requires chemical change	Often physical methods
Example	Copper	Water	Air

CBE Activity - Product Label Investigation

1. Collect safe product labels or packaging.
2. Identify named substances/materials.
3. Classify examples where possible as elements, compounds or mixtures.
4. Record evidence for each classification.
5. Present findings to the class.

THINK BIG Why can the properties of a compound differ greatly from the properties of the elements that form it?

Topical Quiz

6. Define an element.
7. Write the symbols for Na, Cl, Fe and Cu.
8. Why is O₂ not a compound?
9. Distinguish a compound from a mixture.
10. Write a word equation for carbon reacting with oxygen.
11. Classify copper, water, air and salt-sand mixture.
12. Explain why symbol capitalization matters.
13. Give two uses of elements or compounds in daily life.

1.2 Physical and Chemical Changes

Kinetic Theory of Matter

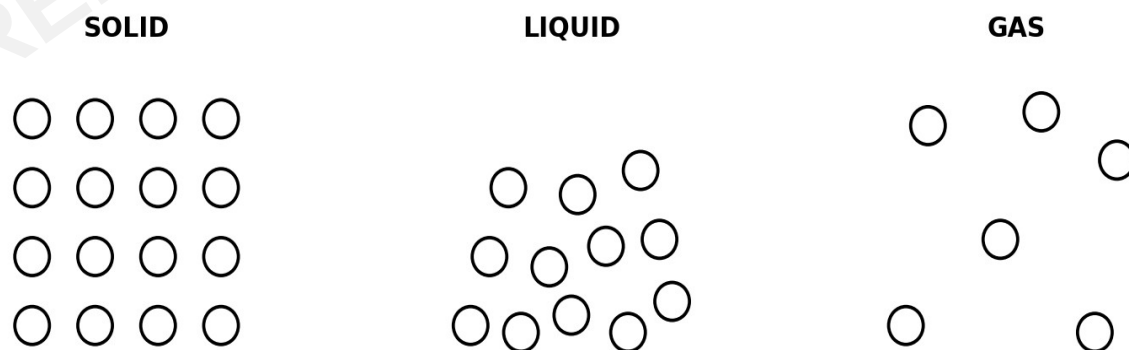


Figure: Particle arrangement in solids, liquids and gases.

The kinetic theory explains matter using tiny particles that are always moving. Heating generally increases particle kinetic energy and motion; cooling reduces it.

Property	Solid	Liquid	Gas
Arrangement	Closely packed and ordered	Close but less ordered	Far apart
Movement	Vibrate about fixed positions	Move past one another	Move freely and rapidly
Shape	Fixed	Takes container shape	Fills container
Volume	Fixed	Fixed	Not fixed
Compression	Very difficult	Difficult	Easy

REMEMBER Heating does not make individual particles swell. Bulk expansion is mainly explained by changes in particle motion and separation.

Changes of State

Change	Direction
Melting	Solid → Liquid
Freezing	Liquid → Solid
Boiling/Evaporation	Liquid → Gas
Condensation	Gas → Liquid

Boiling and Evaporation

Boiling	Evaporation
Occurs throughout the liquid	Occurs at the surface
Occurs at a characteristic boiling temperature under specified pressure	Can occur below boiling point
Bubbles form throughout the liquid	Boiling bubbles are not required

Melting Point, Boiling Point and Heating Curves

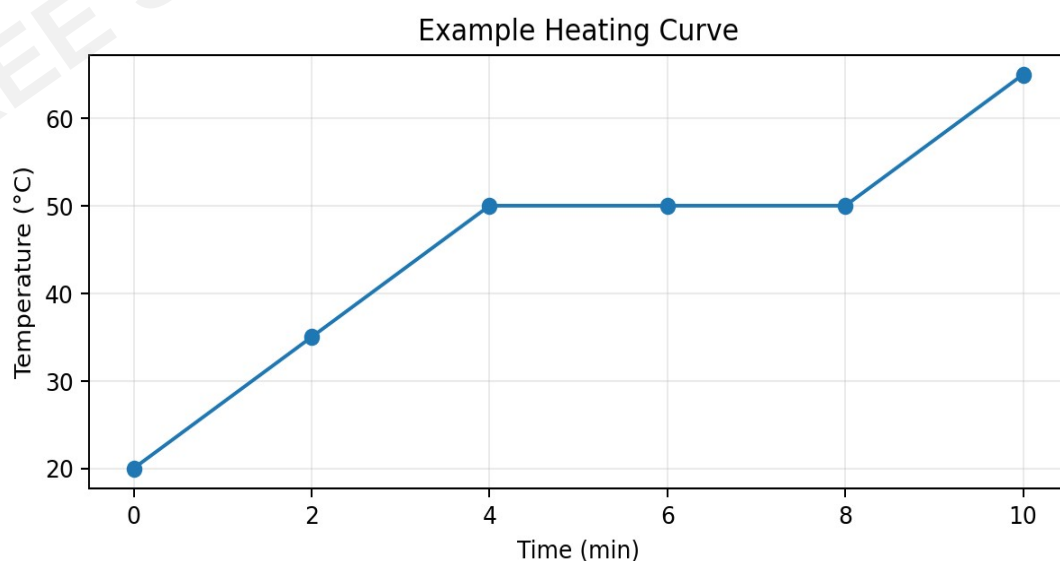


Figure: Example heating curve showing a constant-temperature region.

Light Microscope

Simplified Light Microscope

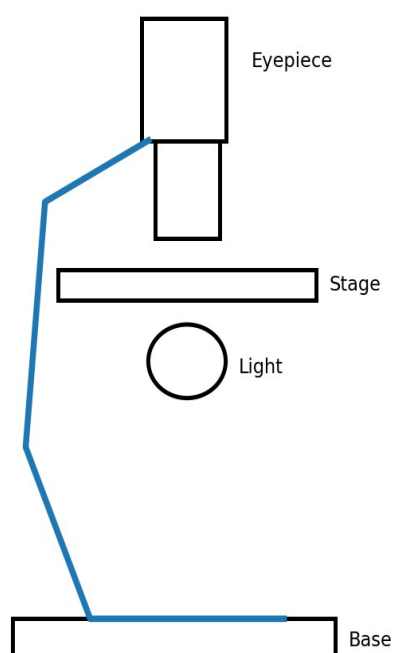


Figure: Simplified light microscope.

Part	Function
Eyepiece	Viewing; provides magnification
Objective lens	Magnifies specimen
Stage	Supports slide
Focus controls	Bring image into focus
Light source/mirror	Illuminates specimen

Total magnification = eyepiece magnification × objective magnification.

Example: $\times 10$ eyepiece $\times \times 40$ objective = $\times 400$.

Practical - Observe Plant Cells

43. Prepare a safe plant-cell specimen under teacher guidance.
44. Place the slide on the stage.
45. Begin with low power.
46. Focus carefully.
47. Observe and draw visible structures using scientific-drawing rules.

Topical Quiz

48. Define a cell.
49. Arrange cell, tissue, organ, organ system and organism.
50. Name three structures common to plant and animal cells.
51. State functions of nucleus, cytoplasm and membrane.
52. Explain why root hair and palisade cells are specialised.

53. Calculate total magnification for $\times 10$ and $\times 20$ lenses.

2.2 Movement of Materials In and Out of the Cell

Diffusion

Diffusion is the net movement of particles from a region of higher concentration to a region of lower concentration.

- Faster particle motion at higher temperature can increase diffusion rate.
- A steeper concentration gradient can increase net diffusion.
- Shorter diffusion distance and larger surface area can support faster exchange.

Osmosis

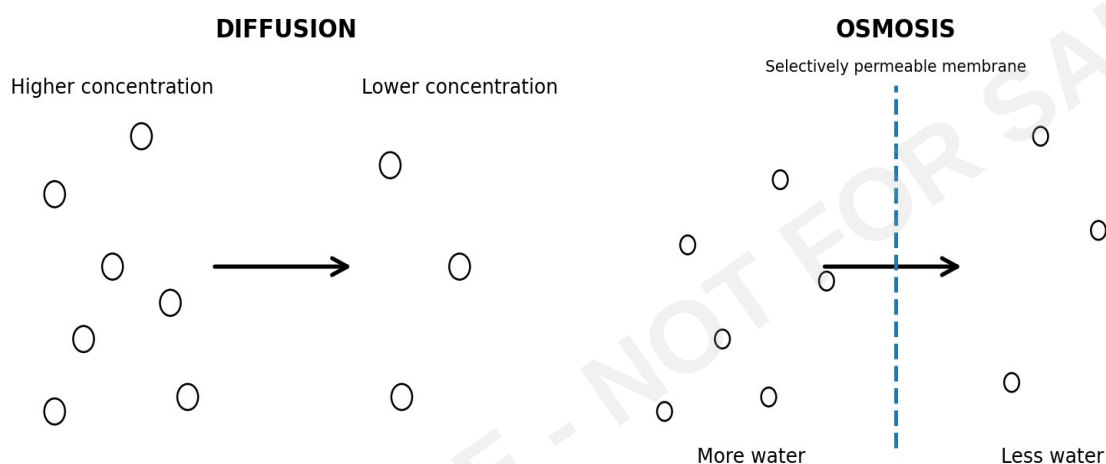


Figure: Diffusion and osmosis compared.

Osmosis is the net movement of water through a selectively permeable membrane from a region of higher water concentration toward a region of lower water concentration.

Diffusion	Osmosis
Can involve different particles	Concerns water
Does not always require a membrane	Requires a selectively permeable membrane
Higher $\rightarrow$ lower particle concentration	Higher $\rightarrow$ lower water concentration

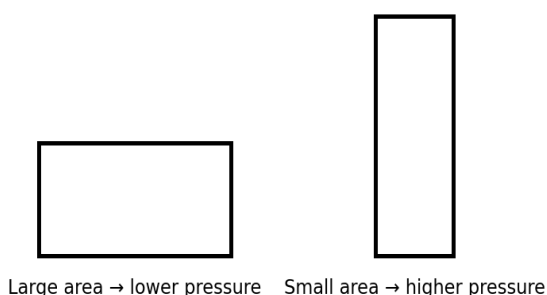
Potato Investigation

Solution	Initial mass (g)	Final mass (g)	Change
A	10	12	+2
B	10	10	0
C	10	7	-3

A gained water, B showed no measurable net change, and C lost water.

Pressure in Solids

CONTACT AREA AND PRESSURE



LIQUID PRESSURE INCREASES WITH DEPTH

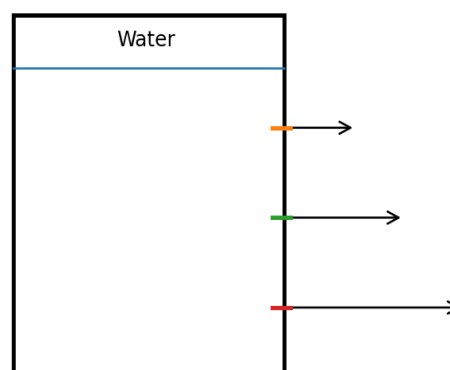


Figure: Contact area and liquid-depth pressure ideas.

For constant force: smaller contact area gives greater pressure; larger area gives lower pressure.

Application	Pressure idea
Sharp knife	Small edge area → high pressure
High heels	Small contact area → high pressure
Broad tyres/tracks	Large area → lower pressure
Building foundation	Large area distributes force

Worked Examples

600 N over 2 m²: $P = 600/2 = 300$ Pa.

2400 N over 0.8 m²: $P = 3000$ Pa.

2400 N over 0.4 m²: $P = 6000$ Pa. Halving area doubles pressure when force is constant.

Pressure in Liquids

Liquid pressure increases with depth. This relationship is treated qualitatively at this level.

In a container with equal holes at different depths, the lowest water jet generally travels farthest because it is at greatest depth.

Everyday Pressure Devices

- Drinking straw: lowering pressure in the straw allows surrounding pressure to help move liquid upward.
- Syringe: pulling the plunger lowers pressure inside; pushing increases pressure and expels fluid.
- Bicycle pump: compression produces a pressure difference that forces air into the tyre.
- Siphon: liquid can flow from a higher level to a lower level through a filled tube under suitable conditions.

SCOPE NOTE Detailed hydraulics and hydraulic-force calculations are not core content in this

PEAK ACHIEVERS ONLINE ACADEMY

GRADE 8 INTEGRATED SCIENCE - BOOK 1

FREE SAMPLE - NOT FOR SALE

READY FOR THE FULL BOOK?

Book 1: **KSh 350**

Book 2: **KSh 450**

BOTH BOOKS: **KSh 550**

This free pack gives you a preview of the actual BOOK 1 learning and revision material.

The complete books contain the full notes, activities, quizzes, mixed revision and model assessment questions.

Contact / WhatsApp: 0740166105 / 0740355410

Email: hezekiahomweri@gmail.com

Website: www.peakachievers.co.ke

After payment, send your confirmation through WhatsApp for PDF delivery.

© 2026 Hezekiah Omweri Nyatuka / Peak Achievers Online Academy.

This sample is for preview and educational evaluation only. It is not for resale.

LEARN • APPLY • THINK BIG • ACHIEVE